

| Coordinate | Descriptor                                      | Unit               | Normalization scale $s_k$ |
|------------|-------------------------------------------------|--------------------|---------------------------|
| $x_1$      | Onboard compute heat                            | J                  | 100                       |
| $x_2$      | Total workload                                  | GFLOP              | 4.8                       |
| $x_3$      | Onboard compute time                            | ms                 | 1400                      |
| $x_4$      | Result transmit time at 50 Mbit s <sup>-1</sup> | ms                 | 100                       |
| $x_5$      | Squared co-location count                       | count <sup>2</sup> | 2401                      |
| $x_6$      | Onboard-result data                             | Mbit               | 4                         |
| $x_7$      | Full-offload transmit heat                      | J                  | 100                       |
| $x_8$      | Ground compute time                             | ms                 | 100                       |
| $x_9$      | Raw input data                                  | Mbit               | 76.8                      |
| $x_{10}$   | Preprocessing heat                              | J                  | 100                       |
| $x_{11}$   | Feature-transmit heat                           | J                  | 100                       |
| $x_{12}$   | Onboard preprocessing work                      | GFLOP              | 2.2                       |
| $x_{13}$   | Remaining ground compute time                   | ms                 | 100                       |
| $x_{14}$   | Total workload descriptor                       | GFLOP              | 4.8                       |
| $x_{15}$   | Collaborative feature data                      | Mbit               | 32                        |

| Quantity                 | Implemented range            | Basis                                                        |
|--------------------------|------------------------------|--------------------------------------------------------------|
| Raw task data            | 8–64 Mbit nominal            | Stress interval reaches 92.8 Mbit before sensitivity scaling |
| Arithmetic workload      | 0.25–4.0 GFLOP nominal       | Stress interval reaches 5.8 GFLOP                            |
| Result / raw-data ratio  | 1–5%                         | Bounded compression assumption                               |
| Feature / raw-data ratio | 8–30%                        | Bounded collaborative-processing assumption                  |
| Preprocessing fraction   | 20–45%                       | Deterministic function of workload quantile                  |
| Onboard / ground rate    | 4 / 80 GFLOP s <sup>−1</sup> | Fixed simulator hardware envelope                            |
| Compute power            | 4–20 W                       | Bracketed by NASA SmallSat avionics examples                 |
| Effective RF return rate | 2.2–220 Mbit s <sup>−1</sup> | NASA SmallSat ground-system examples                         |
| Transmit power           | 6 W                          | Fixed simulator assumption                                   |
| Burst multiplier         | 1.45×                        | Contiguous prespecified stressor                             |

| Regime                | Argmin            | MPC-4             | Bandit             | Threshold         | DQN               | Centered FA       |
|-----------------------|-------------------|-------------------|--------------------|-------------------|-------------------|-------------------|
| Static ( $c = 0$ )    | $24.55 \pm 0.21$  | $24.55 \pm 0.21$  | $24.78 \pm 0.33$   | $24.74 \pm 0.25$  | $26.85 \pm 0.99$  | $26.66 \pm 0.64$  |
| Reduced ( $c = 0.5$ ) | $26.65 \pm 0.63$  | $25.70 \pm 0.26$  | $26.39 \pm 1.05$   | $26.43 \pm 0.38$  | $27.81 \pm 0.68$  | $27.61 \pm 0.56$  |
| Nominal               | $193.42 \pm 8.33$ | $90.83 \pm 1.71$  | $101.00 \pm 4.94$  | $188.08 \pm 6.41$ | $78.33 \pm 1.88$  | $77.66 \pm 1.84$  |
| Burst                 | $241.20 \pm 6.84$ | $108.35 \pm 1.99$ | $136.55 \pm 11.20$ | $238.74 \pm 6.56$ | $97.36 \pm 2.01$  | $96.88 \pm 1.94$  |
| Link-limited          | $272.02 \pm 7.98$ | $115.73 \pm 1.92$ | $139.75 \pm 7.36$  | $227.25 \pm 8.36$ | $102.26 \pm 1.75$ | $101.80 \pm 1.80$ |
| Energy-limited        | $225.31 \pm 7.73$ | $126.00 \pm 1.58$ | $133.26 \pm 4.20$  | $246.46 \pm 6.45$ | $119.67 \pm 2.14$ | $119.00 \pm 1.65$ |
| Thermal-stress        | $247.58 \pm 6.25$ | $109.20 \pm 2.01$ | $137.74 \pm 10.79$ | $241.80 \pm 6.26$ | $97.55 \pm 2.03$  | $97.10 \pm 1.93$  |

| Scope            | Least-favorable DQN objective | Maximum $\Delta$ | 95% interval for maximum | Simultaneous 95% upper bound |
|------------------|-------------------------------|------------------|--------------------------|------------------------------|
| Nominal          | Double DQN                    | -12.43           | [-12.92, -11.81]         | -11.82                       |
| Burst            | Double DQN                    | -10.74           | [-11.26, -10.13]         | -10.17                       |
| Link-limited     | Double DQN                    | -13.15           | [-13.65, -12.58]         | -12.62                       |
| Energy-limited   | Double DQN                    | -5.92            | [-6.66, -4.94]           | -5.04                        |
| Thermal-stress   | Double DQN                    | -11.40           | [-11.94, -10.77]         | -10.84                       |
| All five regimes | Double DQN (Energy-limited)   | -5.92            | [-6.66, -4.94]           | -5.03                        |

| Training objective            | Nominal          | Burst            | Link-limited      | Energy-limited    | Thermal-stress   |
|-------------------------------|------------------|------------------|-------------------|-------------------|------------------|
| Standard DQN                  | $78.33 \pm 1.88$ | $97.36 \pm 2.01$ | $102.26 \pm 1.75$ | $119.67 \pm 2.14$ | $97.55 \pm 2.03$ |
| Full-action Q auxiliary       | $78.14 \pm 2.23$ | $97.21 \pm 2.25$ | $102.17 \pm 2.09$ | $119.78 \pm 2.38$ | $97.41 \pm 2.25$ |
| Immediate advantage auxiliary | $77.84 \pm 1.82$ | $97.22 \pm 2.05$ | $101.95 \pm 1.88$ | $119.17 \pm 2.12$ | $97.40 \pm 2.04$ |
| Centered full-action          | $77.66 \pm 1.84$ | $96.88 \pm 1.94$ | $101.80 \pm 1.80$ | $119.00 \pm 1.65$ | $97.10 \pm 1.93$ |
| Double DQN                    | $78.40 \pm 1.98$ | $97.61 \pm 2.15$ | $102.59 \pm 2.08$ | $120.08 \pm 2.41$ | $97.80 \pm 2.12$ |
| Double + centered full-action | $77.73 \pm 1.99$ | $96.90 \pm 2.02$ | $101.91 \pm 1.88$ | $119.07 \pm 1.77$ | $97.13 \pm 2.01$ |

| Training objective            | Mean cost $\pm$ SD | Matched reference | Paired difference [95% interval] | Holm $p$ | Supported |
|-------------------------------|--------------------|-------------------|----------------------------------|----------|-----------|
| Standard DQN                  | 78.33 $\pm$ 1.88   | —                 | —                                | —        | —         |
| Full-action Q auxiliary       | 78.14 $\pm$ 2.23   | Standard DQN      | -0.19 [-0.56, 0.28]              | 0.05909  | No        |
| Immediate advantage auxiliary | 77.84 $\pm$ 1.82   | Standard DQN      | -0.49 [-0.90, -0.12]             | 1        | No        |
| Centered full-action          | 77.66 $\pm$ 1.84   | Standard DQN      | -0.67 [-0.99, -0.39]             | 0.01288  | Yes       |
| Double DQN                    | 78.40 $\pm$ 1.98   | —                 | —                                | —        | —         |
| Double + centered full-action | 77.73 $\pm$ 1.99   | Double DQN        | -0.67 [-1.18, -0.24]             | 0.03545  | Yes       |

| Regime         | Mean difference | 95% interval   | Negative seeds | Holm $p$ | LOO range      |
|----------------|-----------------|----------------|----------------|----------|----------------|
| Nominal        | -0.67           | [-0.99, -0.39] | 17/20          | 0.01288  | [-0.71, -0.55] |
| Burst          | -0.48           | [-0.73, -0.28] | 17/20          | 0.01288  | [-0.51, -0.37] |
| Link-limited   | -0.46           | [-0.77, -0.17] | 15/20          | 0.08278  | [-0.52, -0.38] |
| Energy-limited | -0.68           | [-1.17, -0.21] | 16/20          | 0.03545  | [-0.79, -0.51] |
| Thermal-stress | -0.45           | [-0.75, -0.22] | 15/20          | 0.08278  | [-0.48, -0.33] |

| Training objective            | Wall time per model (s) | Real interactions | Updates | Modeled action outcomes | Models |
|-------------------------------|-------------------------|-------------------|---------|-------------------------|--------|
| Standard DQN                  | $37.4 \pm 10.4$         | 38,400            | 9,569   | 0                       | 20     |
| Full-action Q auxiliary       | $46.0 \pm 1.4$          | 38,400            | 9,569   | 115,200                 | 20     |
| Immediate advantage auxiliary | $42.9 \pm 1.1$          | 38,400            | 9,569   | 115,200                 | 20     |
| Centered full-action          | $49.0 \pm 3.1$          | 38,400            | 9,569   | 115,200                 | 20     |
| Double DQN                    | $36.0 \pm 1.6$          | 38,400            | 9,569   | 0                       | 20     |
| Double + centered full-action | $53.2 \pm 3.9$          | 38,400            | 9,569   | 115,200                 | 20     |

| Planning horizon | Enumerated sequences | Mean cost | Time per decision (ms) | Held-out traces |
|------------------|----------------------|-----------|------------------------|-----------------|
| 1                | 3                    | 99.33     | 0.13                   | 400             |
| 2                | 9                    | 92.80     | 0.50                   | 400             |
| 4                | 81                   | 90.83     | 4.82                   | 400             |
| 6                | 729                  | 89.35     | 42.50                  | 400             |

| Regime         | $\Delta$ immediate | $\Delta$ penalty | $\Delta$ energy | $\Delta$ latency (s) | $\Delta$ data (Mbit) | Regime         | Low-energy steps DQN/MPC-4 | Min. energy DQN/MPC-4 | Max. heat DQN/MPC-4 |
|----------------|--------------------|------------------|-----------------|----------------------|----------------------|----------------|----------------------------|-----------------------|---------------------|
| Nominal        | 0.76               | -13.26           | -0.104          | -1.04                | -543.9               | Nominal        | 23.6/28.5                  | 0.00/0.00             | 0.192/0.190         |
| Burst          | 0.42               | -11.40           | -0.098          | -0.92                | -489.8               | Burst          | 28.8/32.3                  | 0.00/0.00             | 0.192/0.190         |
| Link-limited   | -0.57              | -12.91           | -0.130          | -1.46                | -388.1               | Link-limited   | 30.2/35.0                  | 0.00/0.00             | 0.193/0.191         |
| Energy-limited | 1.36               | -7.68            | -0.102          | -0.93                | -601.5               | Energy-limited | 38.8/41.8                  | 0.00/0.00             | 0.193/0.190         |
| Thermal-stress | 0.32               | -11.97           | -0.100          | -0.97                | -482.5               | Thermal-stress | 28.8/32.4                  | 0.00/0.00             | 0.364/0.362         |

| Parameter profile      | Mean difference | 95% interval   | Improvement | Holm $p$ |
|------------------------|-----------------|----------------|-------------|----------|
| Reference              | -0.67           | [-0.99, -0.39] | 0.85%       | 0.01804  |
| Compute demand $-20\%$ | -0.49           | [-0.78, -0.22] | 0.79%       | 0.01804  |
| Compute demand $+20\%$ | -0.89           | [-1.35, -0.53] | 0.98%       | 0.00322  |
| Input data $-20\%$     | -1.03           | [-1.55, -0.62] | 1.66%       | 0.01804  |
| Input data $+20\%$     | -0.44           | [-0.69, -0.23] | 0.50%       | 0.01804  |
| Energy use $+20\%$     | -0.60           | [-0.98, -0.28] | 0.57%       | 0.01804  |
| Heat load $+20\%$      | -0.67           | [-1.01, -0.40] | 0.86%       | 0.01804  |
| Link capacity $-20\%$  | -0.64           | [-0.96, -0.35] | 0.72%       | 0.01804  |

| Training preview            | Regime         | Mean difference | 95% interval   | Negative seeds | Holm $p$  | Supported |
|-----------------------------|----------------|-----------------|----------------|----------------|-----------|-----------|
| Simulator-exact preview     | Nominal        | -0.67           | [-0.99, -0.39] | 17/20          | 0.01288   | Yes       |
|                             | Burst          | -0.48           | [-0.73, -0.28] | 17/20          | 0.01288   | Yes       |
|                             | Link-limited   | -0.46           | [-0.77, -0.17] | 15/20          | 0.08278   | No        |
|                             | Energy-limited | -0.68           | [-1.17, -0.21] | 16/20          | 0.03545   | Yes       |
|                             | Thermal-stress | -0.45           | [-0.75, -0.22] | 15/20          | 0.08278   | No        |
| Consequences scaled to 0.85 | Nominal        | -0.45           | [-0.85, -0.12] | 18/20          | 0.002012  | Yes       |
|                             | Burst          | -0.41           | [-0.72, -0.18] | 17/20          | 0.01031   | Yes       |
|                             | Link-limited   | -0.37           | [-0.62, -0.13] | 13/20          | 0.5264    | No        |
|                             | Energy-limited | -0.27           | [-0.71, 0.14]  | 13/20          | 0.5264    | No        |
|                             | Thermal-stress | -0.37           | [-0.70, -0.10] | 16/20          | 0.03545   | Yes       |
| Consequences scaled to 1.15 | Nominal        | -0.77           | [-1.11, -0.49] | 20/20          | 9.537e-06 | Yes       |
|                             | Burst          | -0.48           | [-0.74, -0.29] | 19/20          | 0.0001602 | Yes       |
|                             | Link-limited   | -0.33           | [-0.62, -0.06] | 13/20          | 0.2632    | No        |
|                             | Energy-limited | -0.92           | [-1.45, -0.46] | 17/20          | 0.00773   | Yes       |
|                             | Thermal-stress | -0.42           | [-0.71, -0.18] | 15/20          | 0.08278   | No        |